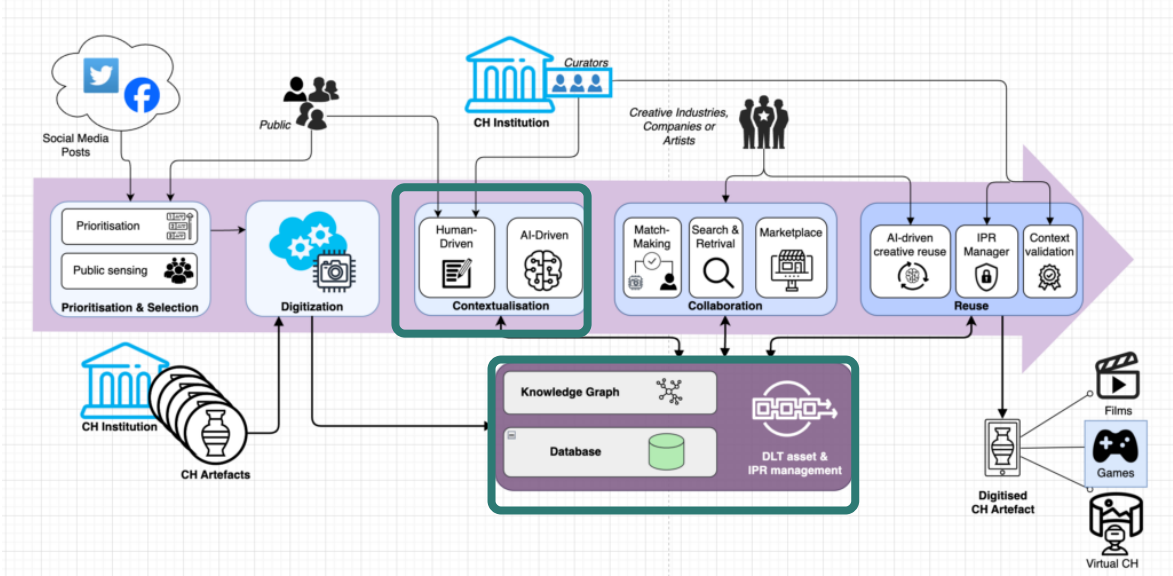
Artwork Type Classification for Metadata Suggestion

Applied Data Science Project · Project design 11/11/2026

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Mentor: LINKS Foundation · Giacomo Blanco

Context: the REEVALUATE project



Source: REEVALUATE project

Where we sit

- **Contextualisation:** a knowledge graph suggests metadata for each new artefact
- **Our classifier** predicts the artefact type from its image
- The **predicted type filters** the metadata taken from the KG

Problem: inconsistent suggestions

Manual annotation is slow

Curators annotate each newly digitised artefact by hand.

Suggestions can be wrong

Without knowing the type, the pipeline may propose metadata that belong to a different kind of artefact.

Today



Suggested material:
“canvas”, “oil paint”

With the predicted type



Materials filtered by type:
textiles only (e.g. silk, cotton)

Value proposition

For **museum cataloguers** who need to **annotate artefacts quickly and accurately**, our **image-based classifier** helps **eliminate nonsensical suggestions** — delivering **faster cataloguing and reliable data for creative reuse** by **predicting the artefact type.**

FOR	WHO NEED	SOLUTION	HELPS	BENEFIT	THROUGH
Museum cataloguers	Quick, accurate annotation	Image-based type classifier	No nonsensical suggestions	Faster cataloguing, reusable data	artefact type prediction

From objective to impact

Objective

The goal we pursue

Classify the artefact type from its image

t(0)

Output

What we produce

An efficient type classifier, ready to be integrated into the pipeline

t(end)

Outcome

The change through use

Fewer wrong metadata proposed to the curator

t(end) + d

Impact

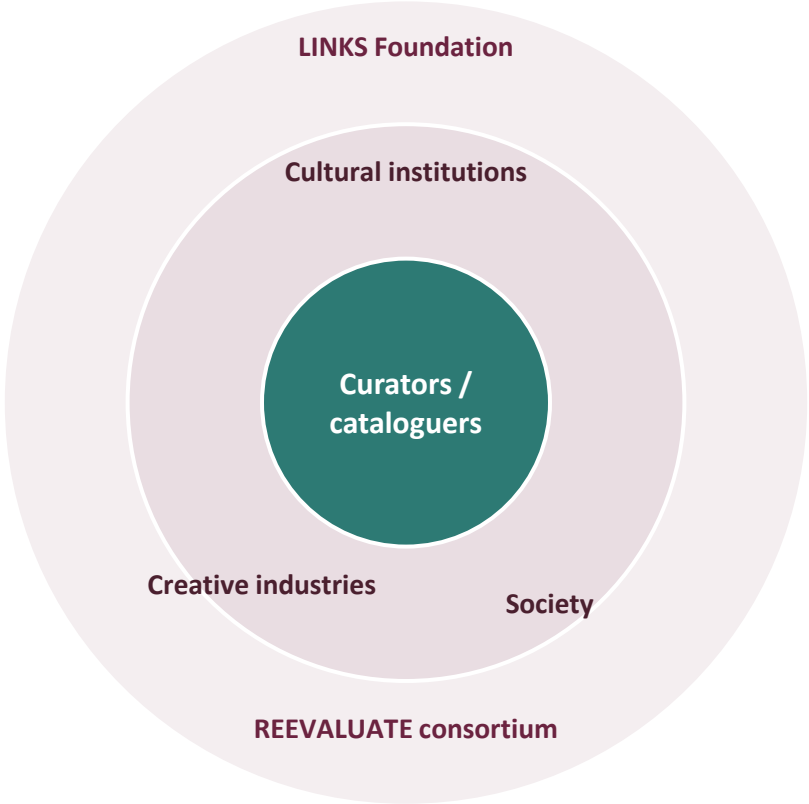
Longer-term effects

Faster, higher-quality digitisation of cultural heritage

t(end) + D

SDG 11.4 Strengthen efforts to protect and safeguard the world's cultural and natural heritage

Stakeholder map



Direct users

Curators and cataloguers who insert artefacts and receive metadata suggestions.

Indirect beneficiaries

Cultural institutions (richer catalogues), creative industries reusing metadata, the public accessing heritage.

Clients & external stakeholders

LINKS Foundation and the REEVALUATE consortium: own the pipeline and the requirements, integrate our module.

Personas



Stefano

Museum Cataloguer

GOAL

Register hundreds of newly digitised artefacts with accurate descriptions, fast

FRUSTRATION

Unrelated suggestions: most of his day goes on deleting them

VALUE OF OUR MODEL

Type recognised at upload, so only relevant options appear



Sara

3D Artist for Games and Films

GOAL

Find real historical clothes and accurate materials in museum collections to recreate them in 3D

FRUSTRATION

“17th-century silk clothing” returns oil paintings of dresses, because garments are tagged “canvas” or “oil paint”

VALUE OF OUR MODEL

Garments classified as “Clothing” get the right materials (silk, cotton)

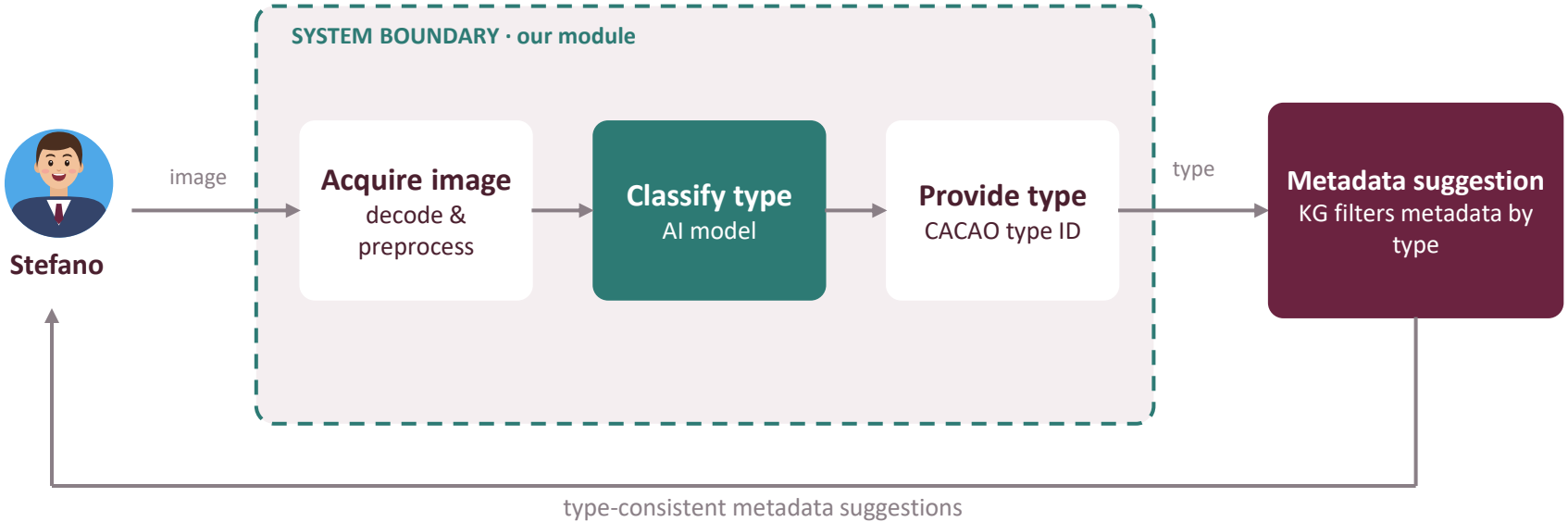
User journey

	1 Upload	2 Receive suggestions	3 Correct	4 Validate
Action by the curator	Uploads the photo of a new artefact	Gets proposed metadata	Fixes wrong fields	Saves the record
Pain point today		Metadata of another kind, e.g. “canvas” for a dress	Time lost removing incoherent values	Risk of wrong data in the catalogue
Opportunity with our module	Predict the type at upload	Filter suggestions by predicted type	Fewer corrections	More reliable catalogue

Functional requirements

ID	Need	Functional requirement
R1	**Know the artefact type**	Given an image of an artefact, the system predicts its type according to the CACAO taxonomy.
R2	**Work on rare types**	The system also recognises types with few examples, not only the dominant ones (Painting, Temples).
R3	**Get the type without waiting**	The system returns the type with low latency, which requires an efficient inference model.

Functional diagram



Data: ARTKB images + CACAO taxonomy

Data: ARTKB

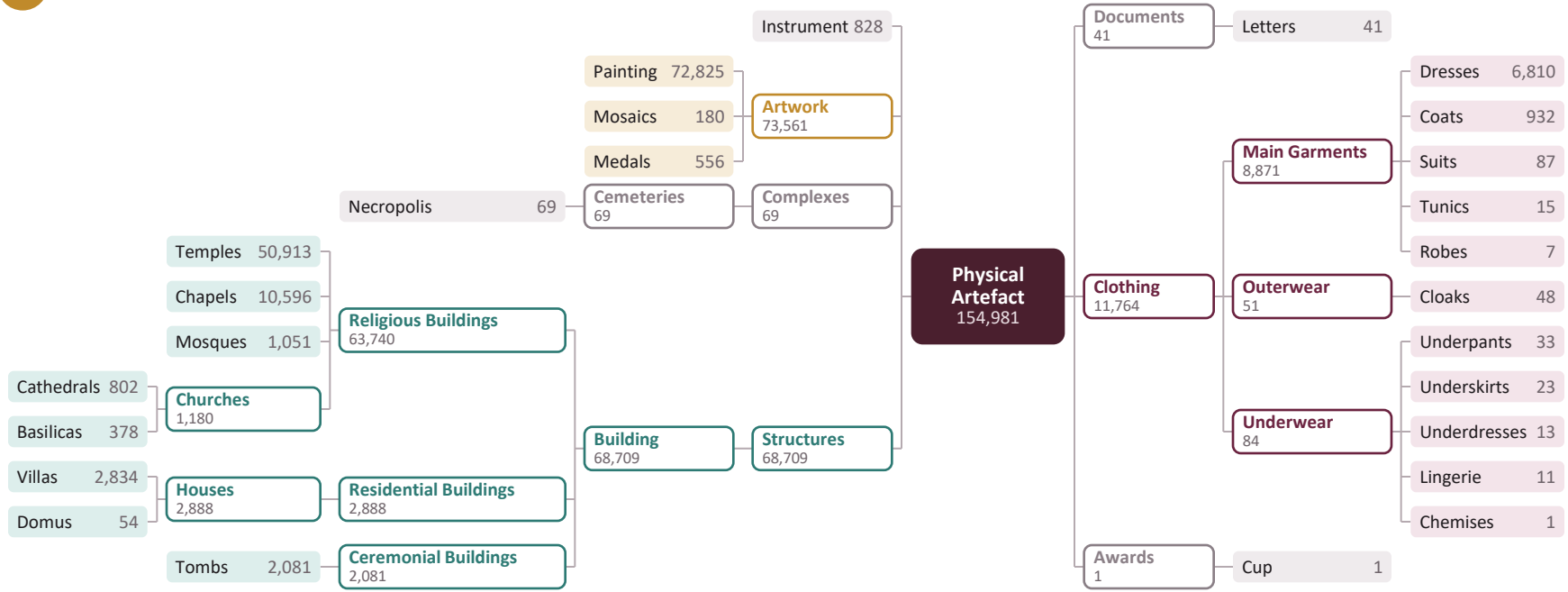
154,981
images, 31 types

80%
in Painting + Temples

13
types with < 100 images

Open & public data (Wikidata)

Working taxonomy to be defined



KPIs

Accuracy

TBD

Share of correctly classified test images

Latency

TBD

Image upload → type, per image, reference hardware

Macro-F1

TBD

F1 averaged over types: every type weighs the same

Top-k accuracy

TBD

True type among the k most likely predictions

Accuracy, macro-F1 and top-k use a fixed, stratified test split. All targets will be set with our mentor after the baseline.

Next steps



1 Fair benchmark

Build a fixed, stratified split on which every experiment is compared

2 Baseline

Data and taxonomy as they are: the reference result

$D_0 + M_0$

3 Data-centric

Tackle class imbalance with data augmentation and vary the dataset taxonomy

$D_1 \ldots D_n + M_0$

4 Model-centric

Combine different architectures with the best taxonomies

$D_i + M_j$

Change one meaningful element at a time, measure the effect on the KPIs, then decide what to keep.

Thank you

Questions? Suggestions?